

Recruiter Project Summary

High-level Executive Portfolio Review

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Recruiter Project Summary

- Business Goal: Automate credit card application risk assessments.
- Problem Statement: Standard classifiers overfit heavily on clean applications, failing to flag defaults due to severe class imbalance (92.5% vs 7.5%).
- Technical Stack: Flask 3.0, SQLite, Scikit-learn (Logistic Regression, Decision Trees, Random Forest), XGBoost, LIME.
- Model Tuning: Evaluated four candidate models. Selected Logistic Regression due to optimal Recall (66.67%) to protect financial assets from default risk.
- Key Features: Secure User/Admin Auth (bcrypt hashing), prediction history logger with CSV/JSON export, interactive admin dashboards, and LIME explainability gauge plots showing individual credit scoring factor attributions.